

Ayudantía 12- MAT1620

1. Utilice multiplicadores de Lagrange para encontrar los valores máximos y mínimos de la función sujeta a la restricción dada:

(a) $f(x, y) = 3x + y; x^2 + y^2 = 10.$

(b) $f(x, y) = 2x + 2y + z; x^2 + y^2 + z^2 = 9.$

2. Mediante multiplicadores de Lagrange, demuestre que el rectángulo de perímetro fijo p es un cuadrado.
3. El plano $x + y + 2z = 2$ intersecta al paraboloide $z = x^2 + y^2$ en una elipse. Encuentre los puntos de esa elipse más cercanos y más lejanos al origen.

1. Utilice multiplicadores de Lagrange para encontrar los valores máximos y mínimos de la función sujeta a la restricción dada:

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$$a) f(x, y) = 3x + y, \quad g(x, y) = x^2 + y^2 = 10$$

$$\nabla f(x, y) = \lambda \cdot \nabla g(x, y)$$

$$g(x, y) = 10$$

$$\nabla f = \langle f_x, f_y \rangle = \langle 3, 1 \rangle$$

$$\nabla g = \langle g_x, g_y \rangle = \langle 2x, 2y \rangle$$

$$\Rightarrow \left. \begin{array}{l} 3 = \lambda \cdot 2x \\ 1 = \lambda \cdot 2y \\ x^2 + y^2 = 10 \end{array} \right\} \quad \frac{3}{1} = \frac{\lambda \cdot 2x}{\lambda \cdot 2y} \Rightarrow 3 = \frac{x}{y}$$

$$\Rightarrow x = 3y$$

$$\hookrightarrow 9y^2 + y^2 = 10 \Rightarrow y^2 = 1 \Rightarrow y = -1 \Rightarrow x = -3$$

$$\Rightarrow y = 1 \Rightarrow x = 3$$

↳ por Lagrange obtenemos los puntos $(-3, -1)$, $(3, 1)$

$$f(-3, -1) = 3 \cdot (-3) + (-1) = -10 \rightarrow \text{mínimo de } f$$

$$f(3, 1) = 3 \cdot 3 + 1 = 10 \rightarrow \text{máximo de } f$$

$$(b) \underbrace{f(x,y,z)}_{f(x,y,z)} = 2x + 2y + z; \underbrace{x^2 + y^2 + z^2}_{g(x,y,z)} = 9.$$

$$\nabla f = \langle f_x, f_y, f_z \rangle = \langle 2, 2, 1 \rangle$$

$$\nabla g = \langle g_x, g_y, g_z \rangle = \langle 2x, 2y, 2z \rangle$$

$$\left. \begin{array}{l} 2 = \lambda \cdot 2x \\ 2 = \lambda \cdot 2y \\ 1 = \lambda \cdot 2z \\ x^2 + y^2 + z^2 = 9 \end{array} \right\} \begin{array}{l} 1 = \frac{x}{y} \Rightarrow x = y \\ 2 = \frac{y}{z} \Rightarrow y = 2z \end{array} \left\{ \begin{array}{l} x = 2z \end{array} \right.$$

$$\begin{aligned} & \hookrightarrow 4z^2 + 4z^2 + z^2 = 9 \Rightarrow z^2 = 1 \Rightarrow z = -1 \quad \nearrow x = y = -2 \\ & \quad \quad \quad \Downarrow \\ & \quad \quad \quad z = 1 \quad \hookrightarrow x = y = 2 \end{aligned}$$

obtenemos los puntos $(-2, -2, -1)$ y $(2, 2, 1)$

$$f(-2, -2, -1) = -9 \quad \rightarrow \text{mínimo}$$

$$f(2, 2, 1) = 9 \quad \rightarrow \text{máximo}$$

de área máxima y

2. Mediante multiplicadores de Lagrange, demuestre que el rectángulo^v de perímetro fijo p es un cuadrado.

$$A(x, y) = x \cdot y, \quad P(x, y) = 2x + 2y = p$$

$$\nabla A = \langle A_x, A_y \rangle = \langle y, x \rangle, \quad \nabla P = \langle P_x, P_y \rangle = \langle 2, 2 \rangle$$

$$\begin{array}{l|l} y = 2\lambda & x = y \\ x = 2\lambda & 2x + 2x = p \\ 2x + 2y = p & \Rightarrow x = \frac{p}{4} \rightarrow y = \frac{p}{4} \end{array}$$

el rectángulo
es un cuadrado

$$A\left(\frac{p}{4}, \frac{p}{4}\right) = \frac{p^2}{16} = \text{Área máxima}$$

$$h(x, y, z)$$

$$\rightarrow x^2 + y^2 - z = 0$$

3. El plano $x + y + 2z = 2$ intersecta al paraboloide $z = x^2 + y^2$ en una elipse. Encuentre los puntos de esa elipse más cercanos y más lejanos al origen.

$$\Rightarrow d(x, y, z) = x^2 + y^2 + z^2$$

$$\hookrightarrow g(x, y, z) = x + y + 2z = 2, \quad h(x, y, z) = x^2 + y^2 - z = 0$$

$$\nabla d = \langle d_x, d_y, d_z \rangle = \langle 2x, 2y, 2z \rangle$$

$$\nabla g = \langle g_x, g_y, g_z \rangle = \langle 1, 1, 2 \rangle$$

$$\nabla h = \langle h_x, h_y, h_z \rangle = \langle 2x, 2y, -1 \rangle$$

$$\begin{array}{l|l} \begin{array}{l} 2x = \lambda + \mu \cdot 2x \\ 2y = \lambda + \mu \cdot 2y \\ 2z = 2\lambda - \mu \\ x + y + 2z = 2 \\ x^2 + y^2 - z = 0 \end{array} & \begin{array}{l} (2x - 2y) = \mu(2x - 2y) \\ \mu(2x - 2y) - (2x - 2y) = 0 \\ (\mu - 1)(2x - 2y) = 0 \\ \hookrightarrow \mu = 1 \\ \hookrightarrow x = y \end{array} \end{array}$$

• Si $\mu = 1$:

$$2z = 2\lambda - 1 \Rightarrow \lambda = \frac{2z + 1}{2}$$

$$\hookrightarrow 2x = \frac{2z+1}{2} + 2x \Rightarrow 2z+1 = 0$$

$$\Rightarrow z = -\frac{1}{2}$$

$$\hookrightarrow \text{como } x^2 + y^2 \geq 0$$

$$\Rightarrow x^2 + y^2 - z = x^2 + y^2 + \frac{1}{2} \geq \frac{1}{2} \neq 0$$

• si $x=y$:

$$\left. \begin{array}{l} 2x + 2z = 2 \\ 2x^2 - z = 0 \end{array} \right\} \begin{array}{l} z = 2x^2 \Rightarrow 2x + 4x^2 = 2 \\ \Rightarrow 2x^2 + x - 1 = 0 \end{array}$$

$$x = \frac{-1 \pm \sqrt{1+8}}{4} = \frac{-1 \pm 3}{4}$$

$$\begin{array}{lcl} \hookrightarrow x = \frac{1}{2} & \circ & x = -1 \\ \Downarrow & & \Downarrow \\ y = \frac{1}{2} & & y = -1 \\ \Downarrow & & \Downarrow \\ z = \frac{1}{2} & & z = 2 \end{array}$$

$$d\left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right) = \frac{1}{4} + \frac{1}{4} + \frac{1}{4} = \frac{3}{4} \quad \rightarrow \text{en el punto } \left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right) \text{ la distancia es mínima}$$

$$d(-1, -1, 2) = 1 + 1 + 4 = 6 \quad \rightarrow \text{en el punto } (-1, -1, 2) \text{ la distancia es máxima}$$

1. Considere $z = f(x, y)$, donde f es una función dos veces diferenciable. Además considere,

$$x(u, v) = 2uv, \quad y(u, v) = u^2 + uv.$$

Calcule

$$\frac{\partial^2 z}{\partial u^2}, \quad \frac{\partial^2 z}{\partial v \partial u}.$$

$$\frac{\partial}{\partial u} \left(\frac{\partial z}{\partial u} \right)$$

$$z = f \begin{cases} x \\ y \end{cases} \begin{cases} u \\ v \end{cases}$$

$$\frac{\partial f}{\partial x} \begin{cases} x \\ y \end{cases} \begin{cases} u \\ v \end{cases}$$

$$\frac{\partial f}{\partial y} \begin{cases} x \\ y \end{cases} \begin{cases} u \\ v \end{cases}$$

$$\frac{\partial z}{\partial u} = \frac{\partial f}{\partial x} \cdot \frac{\partial x}{\partial u} + \frac{\partial f}{\partial y} \cdot \frac{\partial y}{\partial u}$$

$$= \frac{\partial f}{\partial x} \cdot 2v + \frac{\partial f}{\partial y} \cdot (2u + v)$$

$$\frac{\partial^2 z}{\partial u^2} = \frac{\partial}{\partial u} \left(\frac{\partial z}{\partial u} \right) = \frac{\partial}{\partial u} \left(\frac{\partial f}{\partial x} \cdot 2v + \frac{\partial f}{\partial y} \cdot (2u + v) \right)$$

$$= \frac{\partial}{\partial u} \left(\frac{\partial f}{\partial x} \right) \cdot 2v + \frac{\partial}{\partial u} \left(\frac{\partial f}{\partial y} \right) \cdot (2u + v) + \frac{\partial f}{\partial y} \cdot \frac{\partial}{\partial u} (2u + v)$$

$$= \left(\frac{\partial^2 f}{\partial x^2} \cdot \frac{\partial x}{\partial u} + \frac{\partial^2 f}{\partial y \partial x} \cdot \frac{\partial y}{\partial u} \right) 2v + \left(\frac{\partial^2 f}{\partial x \partial y} \cdot \frac{\partial x}{\partial u} + \frac{\partial^2 f}{\partial y^2} \cdot \frac{\partial y}{\partial u} \right) \cdot (2u + v) + 2 \cdot \frac{\partial f}{\partial y}$$

$$\Rightarrow = [f_{xx} \cdot 2v + f_{xy} \cdot (2u + v)] 2v + [f_{yx} \cdot 2v + f_{yy} \cdot (2u + v)] (2u + v) + 2 f_y$$

$$\frac{\partial^2 z}{\partial v \partial u} = \frac{\partial}{\partial v} \left(\frac{\partial z}{\partial u} \right) = \frac{\partial}{\partial v} \left[\frac{\partial f}{\partial x} \cdot 2v + \frac{\partial f}{\partial y} \cdot (2u+v) \right]$$

$$= \frac{\partial}{\partial v} \left(f_x \right) \cdot 2v + f_x \cdot \frac{\partial}{\partial v} (2v) + \frac{\partial}{\partial v} (f_y) \cdot (2u+v) + f_y \cdot \frac{\partial}{\partial v} (2u+v)$$

1. Sea $g : \mathbb{R} \rightarrow \mathbb{R}$ una función derivable tal que $g(0) = -1$ y $g'(0) = 2$. Considere la función $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ definida por:

$$f(x, y) = xy \cdot g\left(\frac{1}{x} - \frac{1}{y}\right)$$

$$\begin{aligned} f(x, y) - z &= 0 \\ F(x, y, z) &= f(x, y) - z \end{aligned}$$

Determine la ecuación del plano tangente a la superficie $z = f(x, y)$ en el punto $(2, 2, f(2, 2))$.

$$\hookrightarrow f_x(x_0, y_0) \cdot (x - x_0) + f_y(x_0, y_0) \cdot (y - y_0) + \underbrace{f(x_0, y_0) - z_0}_{=0} = 0$$

$$\frac{\partial}{\partial x} f = \frac{\partial}{\partial x} \left(xy \cdot g\left(\frac{1}{x} - \frac{1}{y}\right) \right) = y \cdot g\left(\frac{1}{x} - \frac{1}{y}\right) + xy \cdot g'\left(\frac{1}{x} - \frac{1}{y}\right) \cdot \left(-\frac{1}{x^2}\right)$$

$$f_y = x \cdot g\left(\frac{1}{x} - \frac{1}{y}\right) + xy \cdot g'\left(\frac{1}{x} - \frac{1}{y}\right) \cdot \frac{1}{y^2}$$

$$\Rightarrow f_x(2, 2) = 2 \cdot g\left(\frac{1}{2} - \frac{1}{2}\right) + 4 \cdot g'\left(\frac{1}{2} - \frac{1}{2}\right) \cdot \left(-\frac{1}{4}\right)$$

$$= 2 \cdot g(0) + -g'(0) = -2 + -2 = -4$$

$$f_y(2, 2) = 2 \cdot g(0) + g'(0) = -2 + 2 = 0$$

$$f(2, 2) = 4 \cdot g(0) = -4$$

$$z = -4 + -4(x - 2)$$

ecuación plano tg